

Marek Sinagl

Product Engineer + Architect

Iowa City, IA (open to remote)
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Product engineer and architect who builds and operates software end to end, from ambiguous problems through system design, implementation, deployment, and long-term operation. Creator and sole engineer of Archa, a student knowledge-sharing platform in continuous development since 2007, and one of two primary developers co-leading Mingly, a spatial collaboration platform moving from alpha into an institutional pilot. More than ten years of architectural practice brings experience translating complex requirements into practical systems, coordinating across disciplines, and using computational workflows to solve real-world problems.



Two tracks in parallel: shipping software since 2007, practicing architecture since 2011.

Product and software engineering

2023 – present

Co-Lead Software Engineer, Mingly

mingly.world

Spatial collaboration platform for education and multidisciplinary teams using persistent digital rooms.

- Co-lead engineering as one of two primary developers, taking Mingly through alpha toward a fall 2026 pilot at BI Norwegian Business School.
- Built distributed Go/Node.js/WebRTC services and a Nuxt/Vue/TypeScript frontend, using Socket.IO for real-time presence and spatial audio/video.
- Developed shared-room tools for documents, presentations, markup, and collaborative work.

2007 – present

Creator & Sole Software Engineer, Archa

archacvut.cz

Free student platform for organizing and sharing academic materials and experience across generations.

- Created Archa in 2007; today it serves ~20k registered users, with ~6k logging in during the most recent semester and ~3.8M page views.
- Designed a credit-based contribution system for sharing files and feedback on courses and teachers; the platform now contains 345k+ files totaling 551 GB.
- Sole developer and administrator, owning architecture, security, data, deployment, CI/CD, and automation that minimizes manual moderation.
- Modernized the original PHP application to Laravel/PHP 8.5 and Vue/TypeScript while keeping it in production, and built Python/FastAPI workers for extraction, previews, and LLM classification/summarization.

2020 – 2022

Newburyport, MA

Software Engineer, Elyxor Inc.

elyxor.com

- **Packing Slips:** Sole full-stack developer of a warehouse logistics application using Python/Flask, GraphQL, and React; collaborated with QA and a partner on product requirements.
- **Controller:** Built Java/Spring Boot and Vue features for a multi-cloud VM platform spanning AWS and Azure, including power controls, spend monitoring, and shutdown recommendations.

Technical skills

Languages	TypeScript / JavaScript, PHP, Go, Python, Java
Frameworks	Laravel, FastAPI, Vue.js, Nuxt, Node.js, React, Flask, Spring Boot
Data and real-time	MySQL, GraphQL, REST APIs, WebRTC, Socket.IO, background workers, webhooks, file-processing pipelines
Operations	Git, GitHub, CI/CD, Docker, Linux, AWS EC2
AI and automation	LLM APIs, document extraction, classification, summarization
Computational design	Grasshopper, Rhino 3D, Python scripting, Dynamo, Revit

Architecture and design

More than a decade of architectural practice in three countries: understanding what clients need, resolving ambiguous briefs into buildable detail, and coordinating multidisciplinary teams to deliver it.

2022 – present Iowa City, IA	Architectural Designer , Neumann Monson Architects neumannmonson.com • Design and coordinate complex commercial and multifamily projects across clients and multidisciplinary consultant teams, resolving technical requirements into buildable solutions. • Develop BIM and computational workflows using Revit, Grasshopper, Python scripting in Rhino 3D, and Dynamo, alongside building-code analysis and construction documentation.
2015 – 2019 Sydney, Australia	Associate Architect , Silvester Fuller • Project architect, designer, and BIM coordinator on complex residential, mixed-use, and commercial projects.
2011 – 2014 Prague, Czech Republic	Director & Founding Partner , Mio architects • Led architectural projects from concept through construction while managing design, client coordination, and studio operations.

Education and registration

2015	Registered Architect , Czech Chamber of Architects
2011	Master of Architecture & Building Engineering , Czech Technical University in Prague
2009	Bachelor of Architecture & Building Engineering , Czech Technical University in Prague